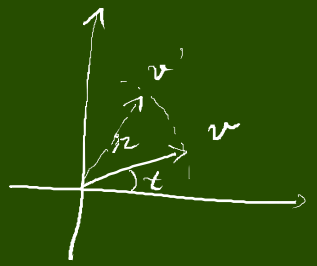
$$v = \begin{pmatrix} x \\ y \end{pmatrix} \longrightarrow v' = \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix}$$

$$\Rightarrow \begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \quad (\text{cond. plane})$$

$$\begin{cases} z = x + iy \\ z = r(\cos t + i\sin t) \end{cases}$$



$$z' = r(\cos(t+\theta) + i\sin(t+\theta))$$

$$= r(\cos t\cos\theta - \sin t\sin\theta) +$$

$$+ ir(\cos t\sin\theta + \sin t\cos\theta)$$

$$z' = x\cos\theta - y\sin t$$

$$+ i(x\sin\theta + y\cos\theta)$$

$$\Rightarrow \underbrace{\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}}_{A} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \longrightarrow \begin{pmatrix} x \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

?

Idem: $A = ?$ proiectia pe $y = \lambda x$, $\lambda \in \mathbb{R}$

$$\begin{pmatrix} x \\ y \end{pmatrix} \longrightarrow \begin{pmatrix} -x \\ y \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Def. $\mathcal{f} : U \longrightarrow V$ **aplic. lin.**

$\Leftrightarrow \forall x, y \in U \quad \forall \lambda, \mu \in K$

$$\mathcal{f}(\lambda x + \mu y) = \lambda \mathcal{f}(x) + \mu \mathcal{f}(y)$$

Exemple. 1) $\mathcal{f} : \mathbb{R}_2[X] \longrightarrow \mathbb{R}_2[X]$ nu e aplic. lin!

$\mathcal{f}(P(x)) = P(X) + 1$

$\mathcal{f}(0) = 1 \neq 0$!

2) $\mathcal{J} : \mathbb{R}_2[X] \longrightarrow \mathbb{R}_2[X]$

$\mathcal{J}(P) = P'$ aplicație liniară.

$P(x) = a + bx + cx^2$

$P'(x) = b + 2cx$

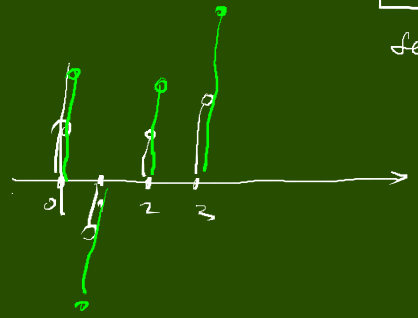
$\mathcal{J}(P+Q) = \mathcal{J}(P) + \mathcal{J}(Q)$

$(P+Q)' = P' + Q'$ ✓

$\mathcal{J}(\lambda P) = \lambda \mathcal{J}(P)$

$(\lambda P)' = \lambda \cdot P'$ ✓

Exemple, 1) $U = \{ \underbrace{(x_n)_{n \in \mathbb{N}}}_{\text{șiruri discrete}} \mid x_n \in \mathbb{R} \}$



$(x_n)_n + (y_n)_n = (x_n + y_n)_n$

$\lambda (x_n)_n = (\lambda x_n)_n$

$\mathcal{L} : U \longrightarrow U$

$\mathcal{L}((x_n)_n) = (0, 0, 0, x_3, x_4, x_5, x_6, 0, \dots)$

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2) $\mathcal{L} : \mathcal{M}_{m,n}(\mathbb{R}) \longrightarrow \mathcal{M}_{m,n}(\mathbb{R})$

$\mathcal{L}(A)_{ij} = \frac{1}{9}\left(a_{i-1,j-1} + \dots + a_{i+1,j+1}\right)$

$A = (a_{ij})$

$\mathcal{L}$ aplic. lin.

$a_{i-1,j+1}$ $a_{i,j+1}$ $a_{i-1,j+1}$
$a_{i,j-1}$ a_{ij} $a_{i,j+1}$
$a_{i+1,j-1}$ $a_{i+1,j}$ $a_{i+1,j+1}$

$\mathcal{L} : U \longrightarrow V$ — aplic. lin.

bij

$\Rightarrow \mathcal{L}^{-1}$ — aplic. lin. ? V

bij ? liniar

Fie $v_1, v_2 \in V$ $\quad \lambda_1, \lambda_2 \in k$

f / $f^{-1}(\lambda_1 v_1 + \lambda_2 v_2) \overset{?}{=} \lambda_1 f^{-1}(v_1) + \lambda_2 f^{-1}(v_2)$?

$f \circ f^{-1} = id_V$

$\lambda_1 v_1 + \lambda_2 v_2 \overset{?}{=} f(\lambda_1 f^{-1}(v_1) + \lambda_2 f^{-1}(v_2))$

$= \lambda_1 f(f^{-1}(v_1)) + \lambda_2 f(f^{-1}(v_2))$

$= \lambda_1 v_1 + \lambda_2 v_2$ ✓

f < bij, proper cont. $\quad \not\Rightarrow \quad f^{-1}$ < bij ✓, proper.

f cont, bij $\overset{?}{\Rightarrow}$ f^{-1} cont

$f : \mathbb{R} \longrightarrow \mathbb{R}$

tema

$U \xrightarrow{\exists! f,g} V$

$\{u_1 \dots u_n\}$ bază $\qquad \{v_1 \dots v_n\} \in V$



Dem. Unicitate : pp. $\exists\, f, g : U \longrightarrow V$ aplic. lin.

$f(u_1) = v_1 \dots f(u_n) = v_n$

$g(u_1) = v_1 \dots g(u_n) = v_n$

$\overset{?}{\Rightarrow} f = g \iff \forall\, u \in U \quad f(u) = g(u)$

Fie $u \in U$

$B = \{u_1 \dots u_n\}$ bază $\subseteq U$ $\Big(\Rightarrow \exists \lambda_1 \dots \lambda_n \in k$

$u = \lambda_1 u_1 + \dots + \lambda_n u_n$

$f(u) = f(\lambda_1 u_1 + \dots + \lambda_n u_n) = \lambda_1 f(u_1) + \dots + \lambda_n f(u_n)$

$= \lambda_1 g(u_1) + \dots + \lambda_n g(u_n) = g(\lambda_1 u_1 + \dots + \lambda_n u_n) = g(u)$

$\Rightarrow \mathcal{J} = g$.

Existența: Fie $u \in U$ $u = \lambda_1 u_1 + \dots + \lambda_n u_n$

$\mathcal{J}(u) = \lambda_1 v_1 + \dots + \lambda_n v_n$

$\begin{cases} \mathcal{J} \text{ bine def.} & V \\ \mathcal{J} \text{ aplic. liu.} \end{cases}$

Exemplu: $U = \mathbb{R}_3[X]$ $V = \mathbb{R}^4$

$\{1, X, X^2, X^3\}$ $\{e_1, e_2, e_3, e_4\}$

bază → bază

$\Rightarrow \exists!$ $\mathcal{J} : \mathbb{R}_3[X] \longrightarrow \mathbb{R}^4$ aplic. liu

$\mathcal{J}(1) = e_1$ $\mathcal{J}(X^2) = e_3$

$\mathcal{J}(X) = e_2$ $\mathcal{J}(X^3) = e_4$

$\mathcal{J}(P) = ?$ $(X-1)^2 \mid P$

$P(1) = 0, P'(1) = 0$

$P(X) = a_0 + a_1 X + a_2 X^2 + a_3 X^3 \longmapsto \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix}$?

$$f(P) = f(a_0 \cdot 1 + a_1 x + a_2 x^2 + a_3 x^3)$$

$$= a_0 f(1) + a_1 f(x) + \dots$$

$$= a_0 e_1 + a_1 e_2 + \dots$$

$$= \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

$$U \xrightarrow{f} V$$

U' $\quad$ V'

$\ker(f)$ $\quad$ $\operatorname{Im}(f)$

$\{u \in U \mid f(u) = 0_V\}$ $\quad$ $\{v \in V \mid \exists u \in U.\ f(u) = v\}$

$f(u) = 0 \Rightarrow u = 0$

f injektivní $\Leftrightarrow$ $\ker(f) = \{0_U\}$

$\Rightarrow$: Je $u \in \ker(f) \Rightarrow f(u) = 0_V$, $f(0) = 0_V$ $\Rightarrow$

f injektivní $\Rightarrow$ $u = 0_U$ $\quad$ $\ker(f) = \{0_U\}$

$\Leftarrow$: Je $u, u' \in U$ a.s. $f(u) = f(u')$

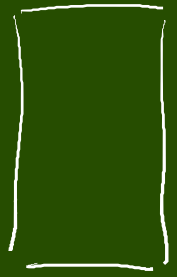
$\Rightarrow f(u) - f(u') = 0$

$f(u - u') = 0$

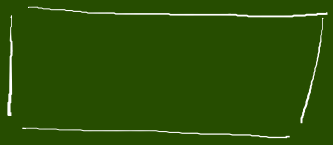
$u - u' \in \ker(f) = \{0_U\}$

$\Rightarrow u - u' = 0_U, \quad u = u'$

$\Rightarrow f$ injektiv



$m \leq n \iff$ Spalten von A sind lin. indep. in K^m



$m \leq n \iff$ Spalten von A erzeugen K^m

Exemplu. $\{1, x, x^2\}$ $\dim_{\mathbb{R}} \mathbb{R}_2[x] = 3$ $\dim_{\mathbb{R}} M_2(\mathbb{R}) = 4$

$$f : \mathbb{R}_2[x] \xrightarrow{\mathbb{R}} M_2(\mathbb{R})$$

$$f(P) = \begin{pmatrix} P(1) & P'(1) \\ 0 & P(1) + P'(1) \end{pmatrix}$$

a) Temă : f aplic. liu.

b) $\ker(f) = ?$ $\quad \operatorname{Im}(f) = ?$

bază, dim.

b) $\ker(f) = \{ P \in \mathbb{R}_2[x] \mid f(P) = 0 \}$

$f(P) = 0 \Rightarrow P(1) = P'(1) = 0$

$P(1) = 0 \Leftrightarrow (x-1) \mid P$

$P(1) = 0, P'(1) = 0 \Leftrightarrow (x-1)^2 \mid P$

$\Rightarrow P(x) = (x-1)^2 \cdot \underbrace{Q(x)}_{\in \mathbb{R}}$

$P(x) = (x-1)^2 \cdot a \quad , \quad a \in \mathbb{R}$

$\ker(f) = \{ a(x-1)^2 \mid a \in \mathbb{R} \} =$

$= Sp \{ (x-1)^2 \}$

$\{ (x-1)^2 \}$ bază în $\ker(f)$

$\dim_{\mathbb{R}}(\ker f) = 1$

$\operatorname{Im}(f) = \{ A \in M_2(\mathbb{R}) \mid \exists P \in \mathbb{R}_2[X] \cdot f(P) = A \}$

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad P = ? \quad P(X) = a_0 + a_1 X + a_2 X^2$

$f(P) = A \quad \Rightarrow \quad \begin{cases} a = a_0 + a_1 + a_2 \\ b = a_1 + 2a_2 \\ c = 0 \\ d = a_0 + 2a_1 + 3a_2 \end{cases}$

$\Rightarrow c = 0$ $\qquad A = \begin{pmatrix} a & b \\ 0 & a+b \end{pmatrix}$

$d = a+b$

$\Rightarrow \begin{cases} a_1 \in \mathbb{R} \\ a_2 = b - a_1 \\ a_0 = a - a_1 - b + a_1 = a - b \end{cases}$

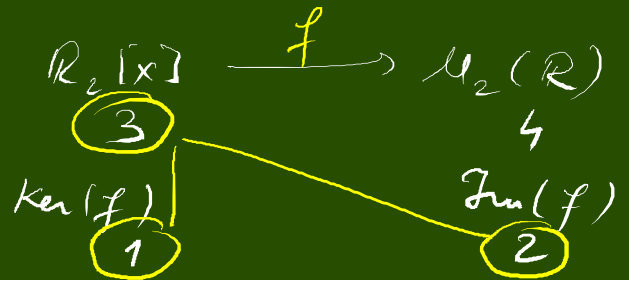
$\mathrm{Im}(f) = \left\{ \begin{pmatrix} a & b \\ 0 & a+b \end{pmatrix} / a, b \in \mathbb{R} \right\} =$

$= \mathrm{Sp}\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \right\}$

lin. indep.

$\Rightarrow \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \right\}$ base' in $\mathrm{Im}(f)$

$\dim_{\mathbb{R}} \mathrm{Im}(f) = 2$



$U \xrightarrow{f} V$

U_1

$\ker(f)$ $\quad \{u_1, \dots u_k\}$ baza în $\ker(f)$

Completăm la $\{u_1, \dots, u_k, u_{k+1}, \dots, u_n\}$ bază în U

$k + \underset{n-k}{?} = n$ $\qquad \downarrow f$

$\{\underbrace{f(u_1)}_{0} \dots \underbrace{f(u_k)}_{0}, f(u_{k+1}) \dots f(u_n)\} \subseteq \operatorname{Im} f,$

$\{\underbrace{f(u_{k+1}), \dots, f(u_n)}_{n-k}\}$

liniar indep $\qquad$ sist. gen. $\operatorname{Im}(f)$

$U \simeq V$

$\exists f : U \longrightarrow V$ izomorfism $\iff \dim_k U = \dim_k V$

$\Rightarrow$ "

$$\dim_K \underbrace{\ker(f)}_{\{0\}} + \dim_K \underbrace{\operatorname{Im}(f)}_{V} = \dim_K U$$

$$0 + \dim_K V = \dim_K U$$

$\Leftarrow$ "

$$\dim_K U = \dim_K V = n$$

bază în U $\quad B = \{u_1, \dots u_n\}$

bază în V $\quad B' = \{v_1, \dots v_n\}$

$\Rightarrow \exists!\ f : U \to V$ aplic. liniară, $f(u_i) = v_i$, $i = \overline{1,n}$

$\exists!\ g : V \to U$ aplic. liniară, $g(v_i) = u_i$, $i = \overline{1,n}$

- $U \xrightarrow{f} V \xrightarrow{g} U$, $g \circ f$: $\quad g \circ f = id_U$

$V \xrightarrow{g} U \xrightarrow{f} V$, $f \circ g$: $\quad f \circ g = id_V$

$g \circ f(u_i) = g(f(u_i)) = g(v_i) = u_i$, $i = \overline{1,n}$

$g \circ f$ aplic. liniară.

id_U aplic. liniară.

$id_U(u_i) = u_i$ $i = \overline{1,n}$

$\Rightarrow f$ bijecție, $g = f^{-1}$

$\Rightarrow g \circ f = id_U$

analog

$f \circ g = id_V$